University of Antwerp
Algorithms and Complexity
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Exercises session 5

Reductions

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1 Definitions

Definition 1.1 (Computable functions) *A function $f : \Sigma^* \to \Sigma^*$ is a* **polynomial time computable function** *if some polynomial time Turing machine exists that halts with just $f(w)$ on its tape, when started with the input $w \in \Sigma^*$.*

Definition 1.2 (Karp reductions) *Language A is* **polynomial time Karp reducible** *to language B, written $A \leq_P B$, if there is a polynomial time computable function $f : \Sigma^* \to \Sigma^*$ such that:*

$$w \in A \iff f(w) \in B.$$

2 Session exercises

1. **Sudoku and Latin Square**
   A **Sudoku** puzzle is a $n^2 \times n^2$ grid that is partially filled with integers, where each integer from 1 to n^2 must appear exactly once in every row, every column, and every $n \times n$ sub-grid. A **Latin Square** of order n is an $n \times n$ grid filled with n different symbols such that each symbol appears exactly once in each row and exactly once in each column. A Latin Square is a special case of a Sudoku grid, where the only constraints are row and column uniqueness, without the additional sub-grid constraints found in Sudoku.
   Task: Prove that Latin Square is as hard as Sudoku.

2. **Course scheduling** Consider a university that needs to schedule courses for a set of students. The goal is to assign each course to a time slot, while ensuring that no student has more than one course scheduled in the same time slot.
   Task: Show how to reduce this problem to a graph colouring problem.

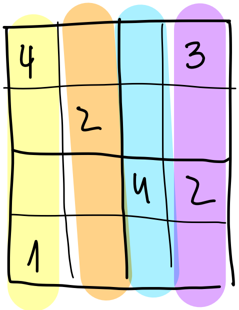
3. **Word Ladder**
   A word ladder involves transforming one word into another by changing one letter at a time, where each intermediate word must also be a valid word in the dictionary. For example, transforming "cat" to "dog" could proceed as follows: "cat" → "cot" → "dot" → "dog".
   Task: Show that this problem can be reduced to the path finding problem in a

graph. Describe the reduction and how finding the shortest path solves the word ladder problem.

4. **Debate tournament**
A university is organizing a debate tournament with teams of three debaters. Each debater has a list of topics they refuse to argue. The goal is to assign debaters to teams so that every team can argue all the topics in the competition. Specifically, each team should have at least one debater who is willing to argue each of the required topics.
Task: Prove that this problem of forming valid debate teams can be reduced to a SAT problem. Show how to encode the debater-topic refusal constraints as a Boolean formula, where satisfying the formula corresponds to finding valid teams for the debate.

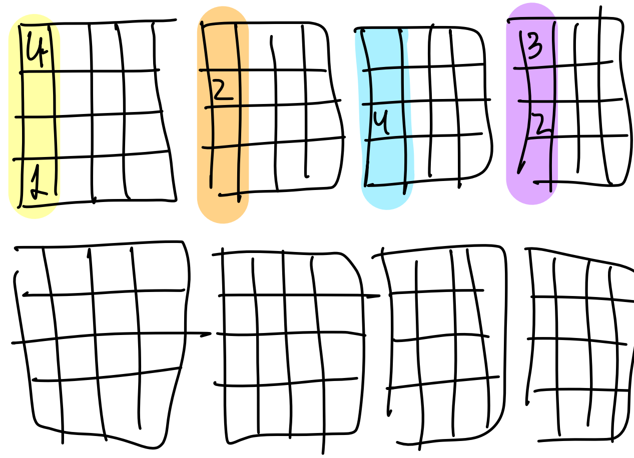
P1] Sudoku & Latin Square

LATIN SQUARE



M_1

SUDOKU



M_2

Fill the blanks

→ with numbers from $n+1$ to n^2

if this can be done in PTIME?

TO PROVE (not hard)

$M_1 \in$ LATIN-SQUARE

$\iff$

$M_2 \in$ SUDOKU

P2 | Course scheduling

① State course scheduling as a decision problem

We have students $S = \{s_1, \dots, s_k\}$ courses $C = \{c_1, \dots, c_\ell\}$ time-slots $T = \{t_1, \dots, t_m\}$

For every student s we define
$\text{courses}(s) = \{c_i : s \text{ has to take course } c_i\}$

Course-scheduling $= \{\langle S, C, \text{courses}(\cdot), \ell \rangle :$ there's a valid course scheduling for students S, with courses assignments $\text{courses}(\cdot)$, in ℓ time-slots$\}$

② Do the reduction

We formulate a graph as follows
$G = (V, E)$
$V = \{c_i : i = 1, \dots, k\}$ → one node for each course
$E = \{(c_i, c_j) : \exists$ a student $s \in S$ such that $c_i, c_j \in \text{courses}(s)\}$ " to avoid schedule-conflicts "

Claim:
there's a valid course scheduling
$\iff$ G is ℓ-colourable (ℓ is the amount of time slots)

How to prove:
* each color corresponds to a time-slot
* nodes (courses) that are connected, cannot have the same color (time slot)

P3 | Word Ladder

① State word ladder as a decision problem

$\text{Word-ladder} = \{ (w_1, w_2) :$ there's a word ladder between w_1 and $w_2 \}$

② Do the reduction

We will define a graph $g = (V, E)$ as follows:

- $V = \{ w : w$ is a word in the dictionary $\}$
- $E = \{ (u, v) : u, v \in V \times V$, u and v differ in 1 letter $\}$

then given the following input for Word Ladder
w_1, w_2
We reduce it to a shortest path problem with instance
(g, w_1, w_2)

③ Prove that the reduction is correct.

We need to prove that
$(w_1, w_2) \in$ Word Ladder $\Longleftrightarrow (g, w_1, w_2) \in$ PATH.

Very straightforward; the sequence of words in word-ladder corresponds to a path in the graph and vice-versa.

P4| Debate tournament

We have topics $T = \{t_1, ..., t_n\}$, debaters $d = \{d_1, ..., d_k\}$ each debater d has $refuse(d) = \{t : d$ refuses to debate topic $t\}$.

We want to know if we can create a team that can cover all topics

- x_d ← debater is in the team
- $x_d \vee \neg x_d^k$ ← is chosen, or doesn't discuss any topic $\forall k$
- $\left(\bigvee_d x_d^k \right)$ ← each topic is covered
- $(\neg x_{d_1} \vee \neg x_{d_2} \vee \neg x_{d_3} \vee \neg x_{d_4})$ ← no more than 3 debaters
  $\forall (d_1, d_2, d_3, d_4) \in V^4$.

Claim: we can form a valid team iff the formula is satisfiable.

TO-DO: Prove the claim (seen in class)